ECET 250E-Linear Circuit 1
Problemset 7-Chapter 8- Second Order RLC Circuits
Solution

Name: _____ Marks: _____

Source Free Series RLC Circuits:

1. **Problem 8.13** For the circuit in Fig. 8.68, calculate the value of *R* needed to have a critically damped response.

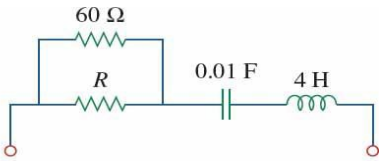


Figure 8.68
For Prob. 8.13.

Chapter 8, Solution 13.

Let R||60 = R_o. For a series RLC circuit,

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.01x4}} = 5 \text{ rads/s}$$

For critical damping, $\omega_o = \alpha = R_o/(2L) = 5$ Nepers/s

or $R_o = 10L = 40 = 60R/(60 + R)$

which leads to $\underline{R = \mathbf{120 \text{ ohms}}}$

2. **Problem 8.17** In the circuit of Fig. 8.71, the switch instantaneously moves from position *A* to *B* at $t = 0$. Find $v(t)$ for all $t \geq 0$.

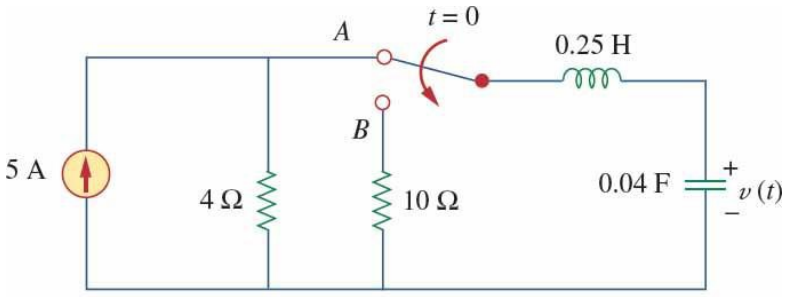


Figure 8.71
For Prob. 8.17.

Chapter 8, Solution 17.

$$i(0) = I_0 = 0,\ v(0) = V_0 = 4x5 = 20$$

$$\frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0) = -4(0+20) = -80$$

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{4}\frac{1}{25}}} = 10$$

$$\alpha = \frac{R}{2L} = \frac{10}{2\frac{1}{4}} = 20,\ which\ is > \omega_o.$$

$$s = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2} = -20 \pm \sqrt{300} = -20 \pm 10\sqrt{3} = -2.679, -37.32$$

$$i(t) = A_1 e^{-2.679t} + A_2 e^{-37.32t}$$

$$i(0) = 0 = A_1 + A_2,\ \frac{di(0)}{dt} = -2.679A_1 - 37.32A_2 = -80$$

$$This\ leads\ to\ A_1 = -2.309 = -A_2$$

$$i(t) = 2.309\left(e^{-37.32t} - e^{-2.679t}\right)$$

$$Since, v(t) = \frac{1}{C}\int_0^t i(t)dt + 20,\ we\ get$$

$$v(t) = \mathbf{[21.55e^{-2.679t} - 1.55e^{-37.32t}]\ V}$$

SOURCE FREE PARALLEL RLC:

3. **Problem 8.23** For the network in Fig. 8.76, what value of C is needed to make the response underdamped with unity damping factor ($\alpha = 1$)?

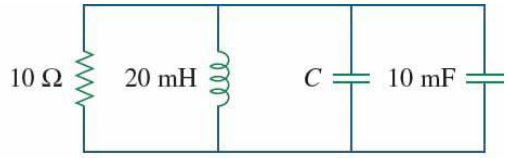


Figure 8.76
For Prob. 8.23.

Chapter 8, Solution 23.

Let $C_o = C + 0.01$. For a parallel RLC circuit,

$$\alpha = 1/(2RC_o),\quad \omega_o = 1/\sqrt{LC_o}$$

$$\alpha = 1 = 1/(2RC_o),\ \text{we then have } C_o = 1/(2R) = 1/20 = 50\ \text{mF}$$

$$\omega_o = 1/\sqrt{0.02x0.05} = 141.42 > \alpha \quad \text{(underdamped)}$$

$$C_o = C + 10\text{ mF} = 50\text{ mF or } C = \mathbf{40\text{ mF}}$$

Step Response of a Series RLC Circuit

4. Problem 8.33 Find $v(t)$ for $t > 0$ in the circuit of Fig. 8.81.

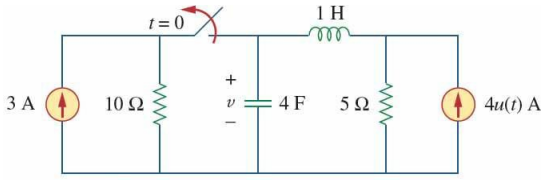
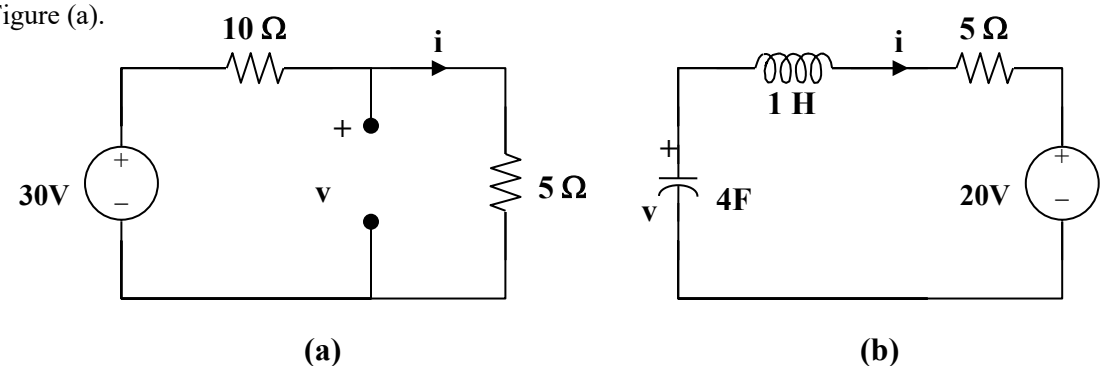


Figure 8.81
For Prob. 8.33.

Chapter 8, Solution 33.

We may transform the current sources to voltage sources. For $t = 0^-$, the equivalent circuit is shown in Figure (a).



$$i(0) = 30/15 = 2\text{ A}, \quad v(0) = 5x30/15 = 10\text{ V}$$

For $t > 0$, we have a series RLC circuit, shown in (b).

$$\alpha = R/(2L) = 5/2 = 2.5$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{4}$$

$$-\alpha \pm \sqrt{\alpha^2 - \omega_o^2} = -2.5 \pm \sqrt{6.25 - 0.25}$$

$$v(t) = V_s + [A_1 e^{-4.949t} + A_2 e^{-0.0505t}], \quad V_s = 20.$$

$$v(0) = 10 = 20 + A_1 + A_2 \text{ or}$$

$$A_2 = -10 - A_1 \qquad (1)$$

$$i(0) = C\,dv(0)/dt \text{ or } dv(0)/dt = -2/4 = -1/2$$

Hence, $$-0.5 = -4.949A_1 - 0.0505A_2 \qquad (2)$$

From (1) and (2), $$-0.5 = -4.949A_1 + 0.0505(10 + A_1) \text{ or}$$

$$-4.898A_1 = -0.5 - 0.505 = -1.005$$

$$A_1 = 0.2052, \quad A_2 = -10.205$$

$$v(t) = \mathbf{[20 + 0.2052e^{-4.949t} - 10.205e^{-0.05t}]\ V}.$$

Step response parallel RLC circuits:

5. **Problem 8.47** Find the output voltage $v_o(t)$ in the circuit of Fig. 8.94.

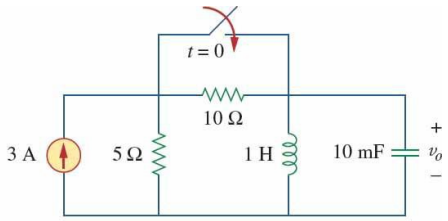


Figure 8.94
For Prob. 8.47.

Chapter 8, Solution 47.

At t = 0-, we obtain, $i_L(0) = 5/(10 + 5) \times 3 = 1A$

and $v_o(0) = 0$.

For t > 0, the 10-ohm resistor is short-circuited and we have a parallel RLC circuit with a step input.

$$\alpha = 1/(2RC) = (1)/(2x5x0.01) = 10$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{1x0.01} = 10$$

Since $\alpha = \omega_o$, we have a critically damped response.

$$s_{1,2} = -10$$

Thus, $i(t) = I_s + [(A + Bt)\, e^{-10t}]$, Is = 3

$$i(0) = 1 = 3 + A \text{ or } A = -2$$

$$v_o = L di/dt = [Be^{-10t}] + [-10(A + Bt)e^{-10t}]$$

$$v_o(0) = 0 = B - 10A \quad \text{or} \quad B = -20$$

Thus, $v_o(t) = \mathbf{(200te^{-10t})\ V}$

6. **Problem 8.49** Determine $i(t)$ for $t > 0$ in the circuit of Fig. 8.96.

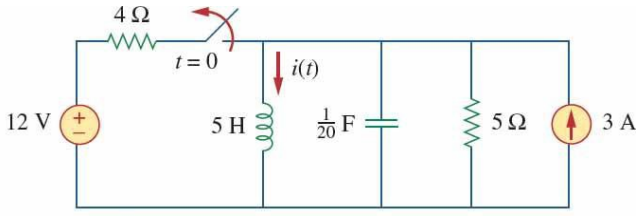


Figure 8.96
For Prob. 8.49.

Solution

For t = 0-, i(0) = 3 + 12/4 = 6 and v(0) = 0.

For t > 0, we have a parallel *RLC* circuit with a step input.

$$\alpha = 1/(2RC) = (1)/(2x5x0.05) = 2$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{5x0.05} = 2$$

Since $\alpha = \omega_o$, we have a critically damped response.

$$s_{1,2} = -2$$

Thus, $i(t) = Is + [(A + Bt)\, e^{-2t}]$, $Is = 3$

$$i(0) = 6 = 3 + A \quad \text{or} \quad A = 3$$

$$v = Ldi/dt \text{ or } v/L = di/dt = [Be^{-2t}] + [-2(A + Bt)e^{-2t}]$$

$$v(0)/L = 0 = di(0)/dt = B - 2x3 \quad \text{or} \quad B = 6$$

Thus, $i(t) = \mathbf{\{3 + [(3 + 6t)e^{-2t}]\}\ A}$

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill